

Engineering Efficient Protein Representations: Identifying the Minimum Effective Dimensionality of ESM-2 Embeddings

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Abstract

Protein language models (PLMs) have introduced high-dimensional sequence representations that can be reused across a wide range of downstream biological machine learning tasks. While these representations can contain substantial biological information, retaining the full embedding dimensionality throughout a production pipeline can impose significant storage, memory, data-transfer, indexing, and downstream computational costs. This work investigates whether the full dimensionality of a protein language model representation is necessary for downstream prediction.

We evaluate embeddings generated using the 650M-parameter ESM-2 model with a native hidden dimension of 1280. Rather than treating the native embedding dimensionality as a fixed engineering requirement, we systematically evaluate principal component analysis (PCA) representations ranging from 32 to 1280 dimensions on protein classification tasks from the TAPE Remote Homology benchmark. The evaluation includes multiple hierarchical biological holdout settings to distinguish dimensionality effects from biological distribution shift.

The results demonstrate a strong diminishing-return relationship between representation dimensionality and downstream performance. For class-level classification under the family-holdout setting, test accuracy increases from 83.73% at 32 dimensions to 97.64% at 256 dimensions and 98.11% at 512 dimensions, reaching 98.27% at 768 dimensions. Increasing dimensionality beyond 768 dimensions provides effectively no additional accuracy benefit in this experiment, despite PCA explained variance increasing from 99.10% at 768 dimensions to 100% at 1280 dimensions. This demonstrates that representation variance and task-relevant information are not equivalent engineering criteria.

The results motivate a practical principle for protein machine learning systems: rather than storing and processing the maximum available representation, systems should identify the smallest representation that satisfies a predefined performance tolerance. We formalize this concept as the *Minimum Effective Representation Dimension* (MERD). For the evaluated class-level Remote Homology task, the results indicate a practical saturation region around 512–768 dimensions. Reducing a 1280-dimensional float32 embedding to 256 dimensions provides an 80% reduction in embedding dimensionality while retaining 93.99% PCA variance and 97.64% classification accuracy on the evaluated family-holdout task.

The work frames embedding dimensionality as an engineering optimization variable rather than merely a representation-learning property. Future experiments will extend the analysis to enzyme-function prediction and additional protein benchmarks, as well as direct measurements of storage, memory bandwidth, vector search, downstream model size, and end-to-end inference cost.

1 Introduction

Protein language models have become an important component of computational biology pipelines. Models such as ESM-2 learn representations from large protein sequence corpora and provide reusable embeddings for downstream tasks including protein classification, structure-related prediction, function prediction, and similarity analysis [1, 2].

The increasing scale of these models has created a practical engineering problem. A downstream system may generate embeddings once and subsequently store millions or billions of protein representations. The representation produced by the model can therefore become a persistent data asset rather than merely an intermediate tensor.

For the ESM-2 650M model evaluated in this work, the native representation has 1280 dimensions. A single float32 representation therefore requires

$$1280 \times 4 = 5120 \text{ bytes} \quad (1)$$

before accounting for metadata, indexing, database overhead, or replication.

Although 5120 bytes is small for an individual protein, the cost scales linearly with the number of proteins. At 100 million proteins, storing the representations as uncompressed float32 arrays requires approximately

$$100,000,000 \times 1280 \times 4 \approx 512 \text{ GB}. \quad (2)$$

A 256-dimensional representation requires only

$$100,000,000 \times 256 \times 4 \approx 102.4 \text{ GB}. \quad (3)$$

This represents a five-fold reduction in raw embedding storage.

The engineering question is therefore not simply whether the original 1280-dimensional representation contains useful information. Instead, the relevant question is:

How many dimensions are actually required for a given downstream biological task?

Using the full representation by default may be unnecessary if downstream performance saturates at a substantially smaller dimensionality.

This work investigates this question empirically.

We define the *Minimum Effective Representation Dimension* (MERD) as the smallest representation dimensionality satisfying a predefined downstream performance tolerance relative to the best available representation:

$$D^* = \min_D \{D : P(D) \geq P_{\max} - \epsilon\}, \quad (4)$$

where $P(D)$ is the performance at dimensionality D , $P_{\max}$ is the best observed performance, and ϵ is an application-defined tolerance.

The purpose of MERD is not to claim that one dimensionality is universally optimal. Instead, it provides an engineering mechanism for determining when additional representation dimensions cease to provide sufficient practical value.

2 Engineering Motivation

The motivation for dimensionality reduction extends beyond disk storage.

For a dataset containing N protein embeddings with dimensionality D and element size b bytes, the raw storage requirement is

$$S = ND b. \quad (5)$$

For float32 embeddings, $b = 4$.

Consequently,

$$\frac{S_{D_2}}{S_{D_1}} = \frac{D_2}{D_1}. \quad (6)$$

This means that reducing dimensionality from 1280 to 256 produces an 80% reduction in raw embedding storage independent of dataset size.

The same linear relationship applies to several downstream operations.

2.1 Storage

Persistent protein embedding repositories, feature stores, vector databases, and model-serving caches all incur costs proportional to embedding dimensionality.

Reducing from 1280 to 256 dimensions produces a factor of

$$\frac{1280}{256} = 5 \quad (7)$$

reduction in raw vector storage.

2.2 Memory

If embeddings are loaded into RAM or GPU memory for batch processing, dimensionality directly determines tensor size:

$$M = NDB. \quad (8)$$

A smaller representation permits larger batches or larger datasets to fit within a fixed memory budget.

2.3 Data Transfer

When embeddings are transmitted between services, dimensionality directly affects the amount of data transferred.

For a distributed architecture in which an embedding service writes vectors to a feature store, reducing D reduces network payload size approximately linearly, assuming the same numeric precision.

2.4 Vector Search

For many dense-vector similarity methods, the computational cost of evaluating a query against a vector is approximately linear in dimensionality:

$$C_{\text{distance}} = O(D). \quad (9)$$

Consequently, reducing the dimensionality of vectors can reduce arithmetic and memory-access requirements for similarity search.

The exact end-to-end benefit depends on the indexing algorithm, hardware, compression method, and implementation, and therefore should be measured rather than assumed.

2.5 Downstream Models

For a linear classifier with K output classes, the number of weights is

$$P = DK. \quad (10)$$

For a downstream multilayer perceptron with hidden width H , the first layer alone contains

$$P_{\text{first}} = DH \quad (11)$$

parameters.

Therefore, reducing the embedding dimension reduces the size of downstream models and their corresponding memory and computational requirements.

2.6 Important Scope of the Current Study

A distinction must be made between *embedding generation* and *embedding consumption*.

PCA applied after ESM-2 inference does not reduce the computational cost of running the original ESM-2 encoder. The encoder still generates the 1280-dimensional representation.

The present work therefore primarily addresses:

- embedding storage,
- memory consumption,
- data movement,
- downstream model size,
- vector operations,
- and downstream inference/search cost.

Encoder-compute reduction requires additional experiments involving smaller PLMs, learned bottlenecks, or architectures capable of producing lower-dimensional representations directly.

3 Research Questions

The study investigates four engineering questions.

1. How does downstream biological prediction performance change as ESM-2 representation dimensionality decreases?
2. At what dimensionality does downstream performance begin to saturate?
3. Does the dimensionality required depend on the biological difficulty of the evaluation task?
4. Can substantial reductions in representation dimensionality be achieved without materially reducing downstream performance?

The central hypothesis is:

The native dimensionality of a protein language model is not necessarily the minimum dimensionality required by downstream systems. For many tasks, substantial reductions in dimensionality can be achieved before meaningful predictive performance is lost.

4 Materials and Methods

4.1 Protein Language Model

Embeddings were generated using ESM-2:

`facebook/esm2_t33_650M_UR50D`

The model contains approximately 650 million parameters and produces a 1280-dimensional hidden representation [1].

The representation used in this study was obtained using masked mean pooling over residue representations. Special beginning-of-sequence and end-of-sequence tokens were excluded from the pooled representation.

The maximum sequence length was set to 512 residues.

4.2 Dataset

The primary benchmark was the TAPE Remote Homology dataset [3].

The dataset contains hierarchical protein labels including:

- class,
- fold,
- superfamily,
- family.

The experiments used:

- 12,312 training proteins,
- 736 validation proteins,
- approximately 1,272–1,284 proteins per test holdout,
- 16,292 unique proteins overall.

The dataset provides multiple biologically structured holdouts, including family, superfamily, and fold holdouts. These splits allow representation dimensionality to be studied under different levels of biological distribution shift.

4.3 Sequence Length

The training sequences had the following length statistics:

Table 1: Training sequence length statistics.

Statistic	Value
Number of sequences	12,312
Mean	167.20
Standard deviation	108.78
Minimum	17
Median	134
75th percentile	217
Maximum	1,419
Sequences > 512 residues	150 (1.22%)

Only approximately 1.22% of training sequences exceeded the 512-residue processing limit, with similarly small proportions in validation and test sets. Therefore, sequence truncation is unlikely to explain the dominant dimensionality trends observed in this experiment.

4.4 Dimensionality Reduction

PCA was fitted exclusively on the training embeddings to prevent information leakage.

The evaluated dimensionalities were:

$$D \in \{32, 64, 128, 256, 512, 768, 1024, 1280\}. \quad (12)$$

For each dimensionality, the same PCA transformation was applied to the training, validation, and test embeddings.

The 1280-dimensional condition represents the unreduced embedding space.

4.5 Downstream Classifier

A multinomial logistic regression classifier was trained for each dimensionality.

This provides a deliberately simple downstream model and allows the experiment to focus on information available in the representation rather than on the capacity of a highly nonlinear classifier.

The main evaluation metrics were:

- accuracy,
- macro-F1,
- balanced accuracy,
- PCA explained variance.

4.6 Hierarchical Evaluation

The experiments evaluated several combinations of target hierarchy and biological holdout.

This distinction is important because a representation can perform well when related biological information is represented in the training distribution while performing substantially worse under more severe distribution shift.

Label overlap was explicitly checked before classification. Test labels that were completely unseen during training were treated as open-set conditions and were not interpreted as conventional closed-set classification results.

5 Experimental Results

5.1 Overview of Experiments

The study was conducted in two stages.

The first stage evaluated the original five-class protein dataset containing approximately 60,000 proteins. This experiment was designed to determine whether the observed classification accuracy of approximately 20% could be explained by representation dimensionality.

The second stage used the TAPE Remote Homology benchmark to determine whether the dimensionality behaviour observed in the first experiment was specific to the original dataset or represented a more general property of protein language model representations.

The complete experimental progression was therefore:

1. learned dimensionality reduction on the original dataset;

2. PCA dimensionality reduction on the original dataset;
3. ESM-2 dimensionality sweep on Remote Homology;
4. hierarchical Remote Homology evaluation;
5. analysis of PCA explained variance versus predictive performance;
6. sequence-length and truncation analysis.

This progression is important because the first experiment alone cannot distinguish between a limitation of the dataset and a limitation of the representation.

5.2 Original Five-Class Dataset: Learned Projection

The initial experiment evaluated a five-class protein classification task:

- Enzima,
- Estrutural,
- Outras,
- Receptora,
- Transporte.

The dataset contained approximately 60,000 proteins and was divided using an 80/20 stratified split, resulting in 48,000 training proteins and 12,000 test proteins.

Because the classes were balanced, random classification corresponds to approximately 20% accuracy.

A learned ESM-2 projection was evaluated across dimensionalities from 64 to 1280.

Table 2: Learned projection dimensionality sweep on the original five-class dataset.

D	Train Accuracy	Test Accuracy	Train Loss
64	0.2074	0.1984	1.6084
128	0.2127	0.2019	1.6083
256	0.2087	0.2002	1.6086
300	0.2114	0.2054	1.6081
360	0.2136	0.2013	1.6083
520	0.2087	0.2036	1.6086
768	0.2063	0.2009	1.6088
1024	0.2107	0.1996	1.6088
1280	0.2048	0.1975	1.6094

The results are notable because increasing dimensionality does not improve performance.

Test accuracy remains between approximately 19.8% and 20.5%, effectively matching the 20% chance baseline.

Training accuracy also remains close to chance, while the final training loss remains close to the expected five-class random cross-entropy value:

$$-\log(1/5) \approx 1.6094. \quad (13)$$

This indicates that the classifier was not simply overfitting the training data. Rather, under this experimental configuration, it failed to learn a meaningful discriminative function from the dataset.

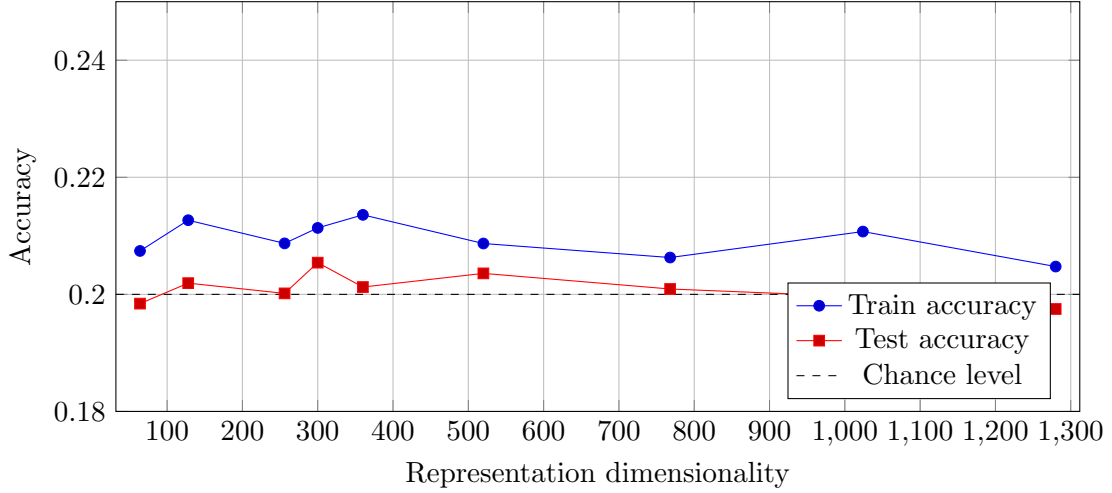


Figure 1: Original five-class learned-projection experiment. Test accuracy remains approximately at the 20% chance level across the entire dimensionality range.

The result initially suggested that dimensionality reduction might not be responsible for the poor classification performance. However, this hypothesis required an independent dataset with a known biological signal.

5.3 Original Five-Class Dataset: PCA

To separate the effect of the learned projection from the effect of dimensionality reduction itself, PCA was subsequently applied to the same original ESM-2 embeddings.

Table 3: PCA dimensionality sweep on the original five-class dataset.

D	Train Accuracy	Test Accuracy	Explained Variance
64	0.2061	0.2023	0.9166
128	0.2135	0.1953	0.9521
256	0.2248	0.1983	0.9738
300	0.2237	0.2018	0.9778
360	0.2260	0.2003	0.9820
520	0.2309	0.2028	0.9893
768	0.2339	0.2013	0.9953
1024	0.2386	0.1998	0.9985
1280	0.2420	0.2047	1.0000

The PCA experiment produces the same qualitative result.

Although PCA explained variance increases from approximately 91.66% at 64 dimensions to 100% at 1280 dimensions, test accuracy remains approximately 20%.

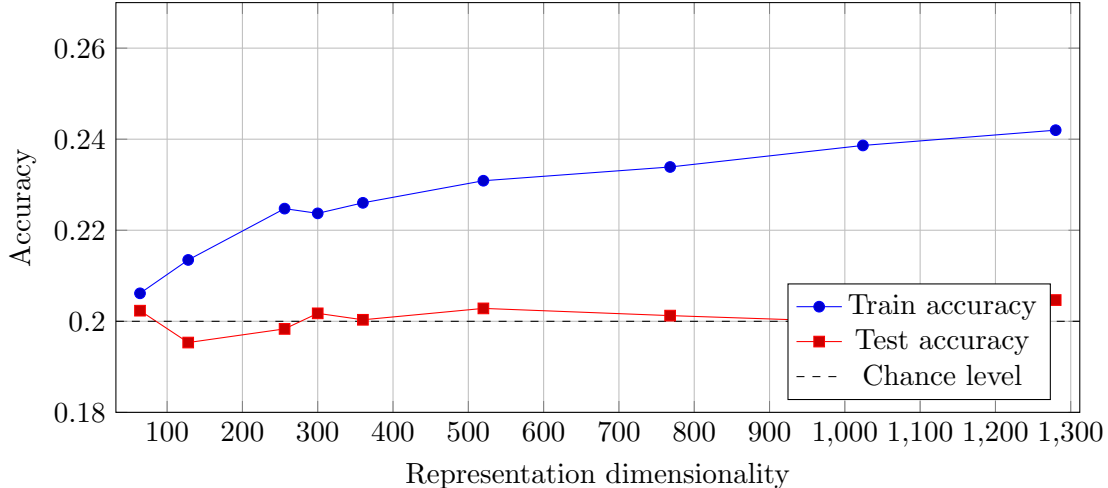


Figure 2: Original five-class PCA experiment. Despite increasing explained variance with dimensionality, test accuracy remains close to the 20% chance level.

The combination of the learned projection and PCA experiments provides strong evidence that the approximately 20% result is not simply caused by PCA compression.

5.4 Remote Homology: A Strong Dimensionality-Dependent Signal

The TAPE Remote Homology benchmark produced substantially different behaviour.

For class-level classification on the family-holdout set, test accuracy increased strongly with representation dimensionality.

Table 4: ESM-2 dimensionality sweep for class-level Remote Homology classification under family holdout.

D	Train Acc.	Valid Acc.	Test Acc.	Test Macro-F1	Explained Var.
32	0.8002	0.7391	0.8373	0.6241	0.7676
64	0.8574	0.7880	0.9009	0.8220	0.8326
128	0.9207	0.8451	0.9607	0.9147	0.8906
256	0.9497	0.8899	0.9764	0.9709	0.9399
512	0.9638	0.9049	0.9811	0.9789	0.9770
768	0.9677	0.9076	0.9827	0.9798	0.9910
1024	0.9704	0.9076	0.9827	0.9799	0.9973
1280	0.9709	0.9090	0.9827	0.9799	1.0000

This experiment demonstrates a fundamentally different behaviour from the original five-class dataset.

At 32 dimensions, test accuracy is 83.73%.

At 256 dimensions, it increases to 97.64%.

At 512 dimensions, it reaches 98.11%.

At 768 dimensions, it reaches 98.27%.

No further improvement is observed at 1024 or 1280 dimensions.

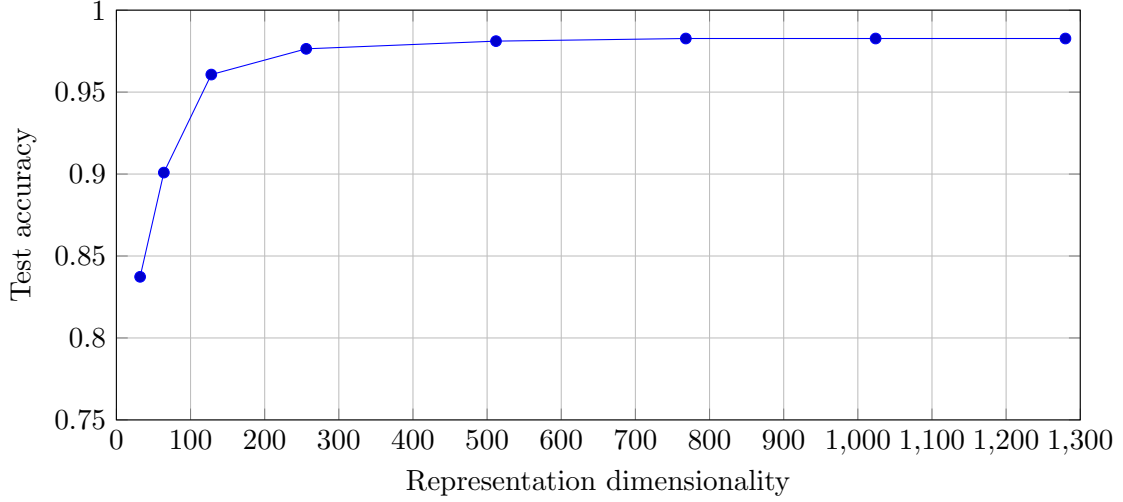


Figure 3: Remote Homology class-level accuracy as a function of representation dimensionality. Performance increases rapidly at low dimensions and enters a saturation regime around 512–768 dimensions.

The curve demonstrates the central engineering phenomenon investigated in this study: representation dimensionality has a strong effect at low dimensions but diminishing downstream returns at higher dimensions.

5.5 Macro-F1 Across Dimensionality

Accuracy alone can hide class imbalance effects. Macro-F1 therefore provides an additional measure of whether performance improvements occur across classes.

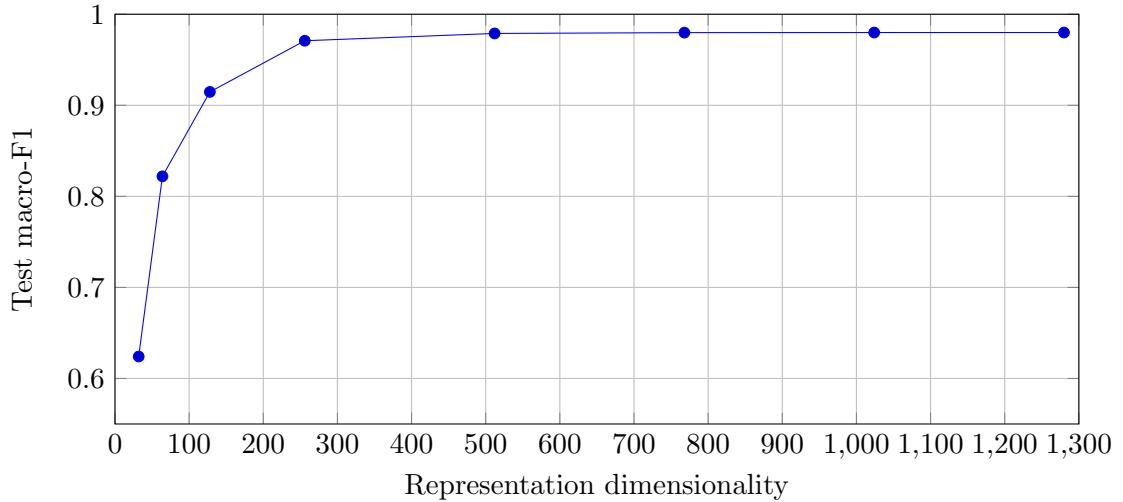


Figure 4: Macro-F1 exhibits the same saturation behaviour as accuracy, with a particularly large improvement between 32 and 256 dimensions.

The increase in macro-F1 from 0.6241 at 32 dimensions to 0.9709 at 256 dimensions is particularly important. It indicates that the dimensionality effect is not driven solely by performance on the largest classes.

5.6 Explained Variance Versus Predictive Utility

The relationship between PCA explained variance and downstream performance is one of the most important observations in the experiment.

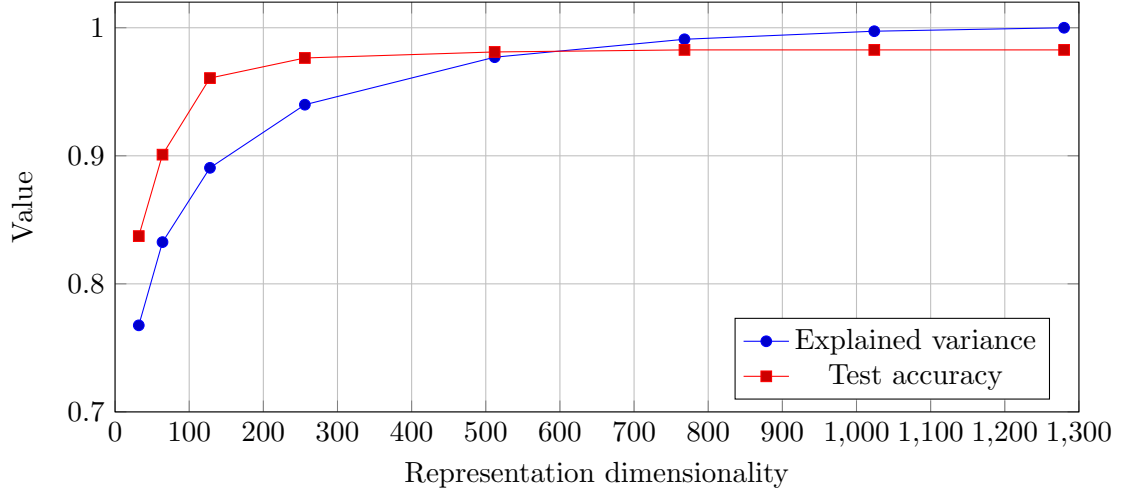


Figure 5: PCA explained variance continues increasing toward 100%, whereas downstream classification accuracy saturates much earlier. Statistical variance preservation and task-relevant information preservation are therefore not equivalent.

At 256 dimensions, 93.99% of the PCA variance is retained while test accuracy reaches 97.64%.

At 768 dimensions, explained variance increases to 99.10%, but test accuracy increases by only 0.63 percentage points relative to 256 dimensions.

This provides direct empirical evidence for diminishing returns.

5.7 Biological Holdout Difficulty

The effect of dimensionality must be interpreted together with biological distribution shift.

Table 5: Best observed class-level performance across Remote Homology biological holdouts.

Holdout	Best D	Test Accuracy	Test Macro-F1
Family	768	0.9827	0.9798
Superfamily	1280	0.9282	0.8718
Fold	768	0.8900	0.7602

The family-holdout task reaches 98.27% accuracy, while the superfamily and fold holdouts reach substantially lower performance.

This demonstrates that increasing representation dimensionality cannot completely compensate for biological distribution shift.

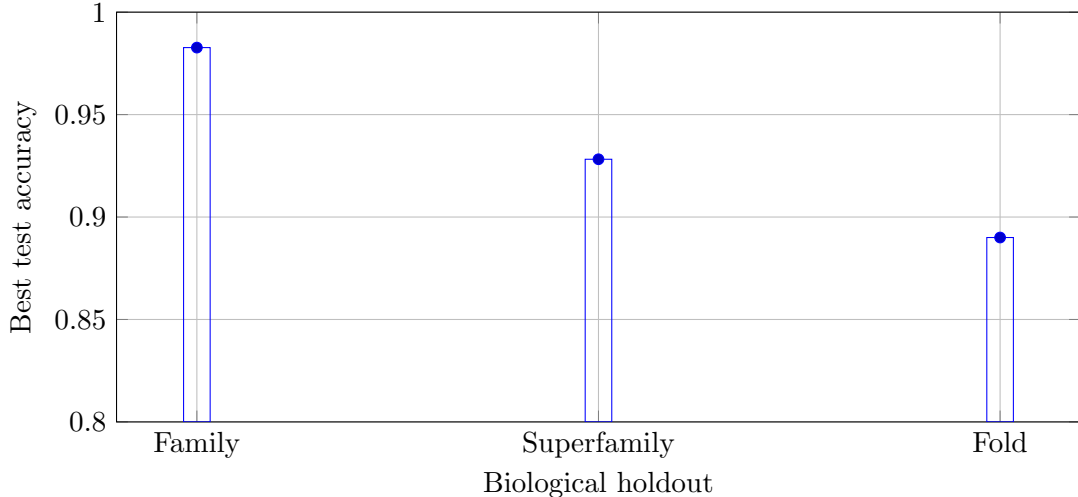


Figure 6: Best class-level test accuracy decreases as the biological generalization problem becomes more difficult. The difference persists despite access to high-dimensional ESM-2 representations.

This result is important from an engineering perspective because it shows that simply allocating more memory to the embedding does not guarantee better generalization.

5.8 Hierarchical Classification Results

The Remote Homology hierarchy was also evaluated at multiple label levels.

Table 6: Selected hierarchical classification results.

Target	Holdout	Best D	Accuracy	Macro-F1	Balanced Acc.
Class	Family	768	0.9827	0.9798	—
Class	Superfamily	1280	0.9282	0.8718	—
Class	Fold	768	0.8900	0.7602	—
Fold	Family	768	0.9796	0.9497	0.9603
Fold	Fold	1024	0.3162	0.1286	0.1778
Family	Family	1024	0.9623	0.9146	0.9374

The results show that the dimensionality required for effective prediction depends on both the biological target and the generalization regime.

For example, family-level classification under the family-holdout condition reaches 96.23% accuracy at 1024 dimensions, while fold-level classification under the fold-holdout condition achieves only 31.62% accuracy at its best observed dimensionality.

These results should not be interpreted as evidence that larger representations are universally superior. Instead, they demonstrate that task complexity and distribution shift determine the amount of representation capacity that is useful.

5.9 Sequence Length Analysis

Sequence truncation was evaluated as a potential confounding factor.

The training sequence statistics were:

Table 7: Remote Homology training sequence-length statistics.

Statistic	Value
Number of sequences	12,312
Mean length	167.20
Standard deviation	108.78
Minimum	17
Median	134
75th percentile	217
Maximum	1,419
Sequences > 512	150 (1.22%)

Only 1.22% of training sequences exceeded the 512-residue processing limit. The corresponding proportions were 1.49% for validation and 0.79% for the evaluated test set.

Consequently, sequence truncation is unlikely to account for the dominant dimensionality-performance relationship observed in the Remote Homology experiments.

6 Engineering Interpretation of the Results

The experiments establish three distinct regimes.

6.1 Regime I: Excessive Compression

At very small dimensions, such as 32 and 64, substantial predictive information is lost.

For the Remote Homology family-holdout task, accuracy increases from 83.73% at 32 dimensions to 90.09% at 64 dimensions and 96.07% at 128 dimensions.

Therefore, aggressive compression can be harmful.

6.2 Regime II: Efficient Representation

Between approximately 128 and 512 dimensions, additional dimensions produce substantial performance improvements.

At 256 dimensions, the classifier already achieves 97.64% accuracy.

This represents an 80% reduction in representation dimensionality relative to the native 1280-dimensional ESM-2 representation.

6.3 Regime III: Diminishing Returns

Beyond approximately 512–768 dimensions, additional dimensions produce negligible improvements for the evaluated task.

Accuracy changes from 98.11% at 512 dimensions to 98.27% at 768 dimensions and remains at 98.27% at 1024 and 1280 dimensions.

At the same time, storage continues to increase linearly.

This creates the engineering optimization problem addressed by this work:

$$\max_D P(D) \tag{14}$$

is not equivalent to

$$\min_D C(D) \quad \text{subject to} \quad P(D) \geq P_{\text{target}}. \tag{15}$$

The second formulation is the more appropriate objective for production systems.

7 Representation Cost Analysis

For a protein embedding with D float32 dimensions, the storage cost is

$$S_{\text{protein}} = 4D \text{ bytes.} \quad (16)$$

The resulting storage requirements for 100 million proteins are shown below.

Table 8: Raw float32 storage requirements for 100 million protein embeddings.

D	Bytes/protein	Storage	Reduction vs 1280D
1280	5,120	512.0 GB	0%
1024	4,096	409.6 GB	20%
768	3,072	307.2 GB	40%
512	2,048	204.8 GB	60%
256	1,024	102.4 GB	80%
128	512	51.2 GB	90%
64	256	25.6 GB	95%
32	128	12.8 GB	97.5%

The relationship is linear:

$$S(D) \propto D. \quad (17)$$

However, the observed classification curve is not linear.
This mismatch is the core engineering opportunity.

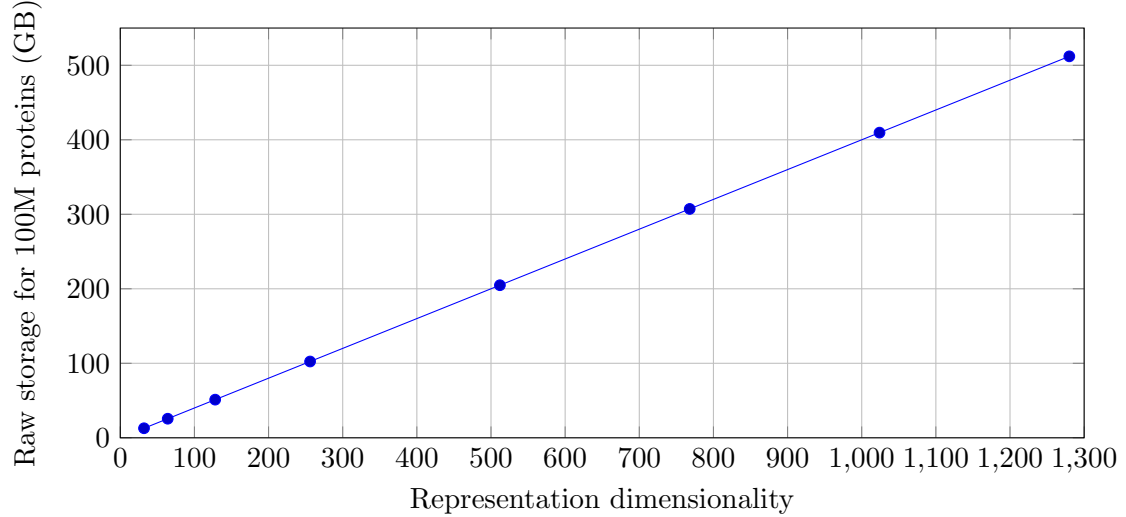


Figure 7: Raw float32 storage scales linearly with embedding dimensionality. At 100 million proteins, reducing from 1280D to 256D reduces the representation footprint from approximately 512 GB to 102.4 GB.

8 Minimum Effective Representation Dimension

The experimental results motivate the definition of the Minimum Effective Representation Dimension (MERD).

Given a downstream performance metric $P(D)$ and acceptable tolerance ϵ :

$$D^* = \min_D \{D : P(D) \geq P_{\max} - \epsilon\}. \quad (18)$$

For the class-level Remote Homology family-holdout experiment:

$$P_{\max} = 0.9827. \quad (19)$$

The difference between 256D and the maximum observed performance is:

$$0.9827 - 0.9764 = 0.0063. \quad (20)$$

Thus, 256D is within approximately 0.63 percentage points of the best observed accuracy while reducing dimensionality by 80%.

At 512D:

$$0.9827 - 0.9811 = 0.0016. \quad (21)$$

Therefore, if an application permits a 0.2 percentage-point tolerance, 512D would satisfy the criterion in this experiment.

This demonstrates why MERD should be defined relative to an application-specific tolerance rather than treated as a universal constant.

9 Key Findings

The complete set of experiments supports the following engineering findings.

1. **The original 20% classification result is not explained by PCA compression alone.**

Both learned projection and PCA experiments remained near chance on the original five-class dataset across the dimensionality range.

2. **ESM-2 dimensionality can strongly affect downstream performance when the task contains a learnable biological signal.**

Remote Homology accuracy increased from 83.73% at 32 dimensions to 98.27% at 768 dimensions.

3. **Downstream performance exhibits diminishing returns.**

For class-level family-holdout classification, performance effectively saturates around 512–768 dimensions.

4. **Explained variance is not equivalent to downstream utility.**

PCA variance increases from 93.99% at 256 dimensions to 100% at 1280 dimensions, while classification accuracy changes only marginally.

5. **Biological distribution shift can dominate dimensionality.**

The best class-level performance decreases from 98.27% under family holdout to 92.82% under superfamily holdout and 89.00% under fold holdout.

6. **Smaller representations can provide substantial infrastructure savings.**

Reducing from 1280D to 256D produces an 80% reduction in raw embedding storage and a five-fold reduction in vector dimensionality.

7. **There is no universally optimal dimensionality.**

The appropriate dimensionality depends on the downstream task, performance tolerance, and biological generalization regime.

10 Engineering Design Principle

The combined results motivate the following production design principle:

Do not automatically retain the maximum-dimensional representation produced by a protein language model. Measure the downstream performance–cost curve and deploy the smallest representation that satisfies the application’s performance requirement.

The practical optimization target is therefore:

$$\boxed{\min_D C(D) \quad \text{subject to} \quad P(D) \geq P_{\text{target}}} \quad (22)$$

where $C(D)$ represents the total operational cost associated with the representation.

This formulation changes dimensionality reduction from a purely statistical preprocessing step into an explicit machine learning systems optimization problem.

11 Discussion

The most important observation from the experiments is that representation dimensionality should be treated as an engineering resource.

The native 1280-dimensional ESM-2 representation is useful because it provides a rich representation space. However, a downstream application does not necessarily need to retain every available dimension.

The Remote Homology experiment demonstrates that a large fraction of the useful predictive signal can be retained in a much smaller representation. At 256 dimensions, 80% of the original dimensions can be removed while retaining 97.64% test accuracy on the evaluated class-level task.

The benefit of reducing dimensionality becomes increasingly important as the number of proteins increases. At 100 million proteins, the difference between 1280D and 256D is approximately 409.6 GB of raw float32 storage.

The same reduction affects memory requirements and the size of vectors passed between system components. For downstream linear and neural models, the input dimensionality also directly influences parameter counts and computation.

However, the experiments also demonstrate why dimensionality should not be minimized blindly.

At 32 dimensions, the Remote Homology classifier loses substantial predictive performance. The goal is therefore not to find the smallest possible representation, but the smallest representation that remains sufficiently useful.

This motivates the MERD framework.

An engineering system can specify a tolerance such as:

$$\epsilon = 0.01 \quad (23)$$

and select the smallest D whose performance is within one percentage point of the best observed model.

This creates a measurable trade-off between:

$$\text{accuracy} \leftrightarrow \text{storage} \leftrightarrow \text{memory} \leftrightarrow \text{data movement} \leftrightarrow \text{search cost}. \quad (24)$$

The appropriate operating point can therefore be selected according to the requirements of the production system.

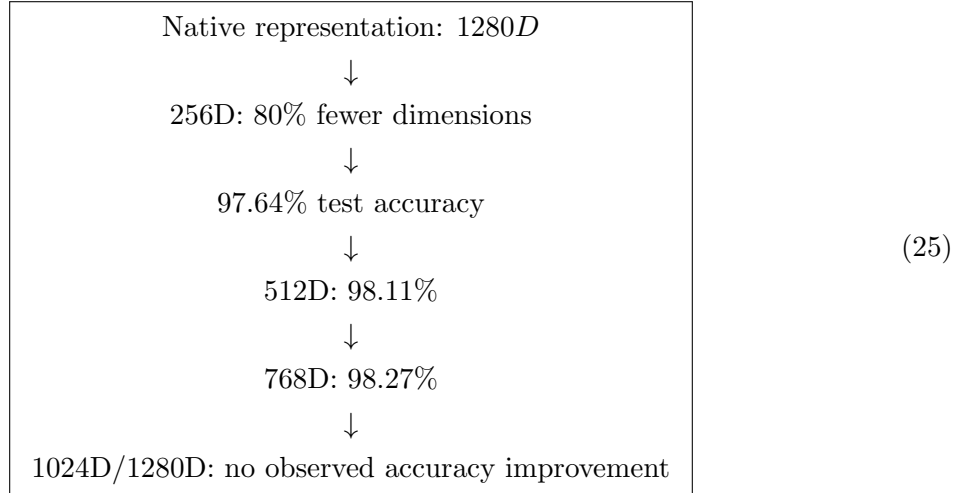
Importantly, the present work does not claim that post-hoc PCA reduces the computational cost of generating ESM-2 embeddings. The ESM-2 encoder still produces the original 1280-dimensional representation before PCA is applied.

Instead, the demonstrated savings occur downstream of the encoder.

Future experiments involving smaller language models or learned bottlenecks are required to determine whether similar principles can reduce the cost of representation generation itself.

12 Summary of Experimental Evidence

The experimental evidence can be summarized as follows:



The engineering implication is straightforward:

Once downstream performance has reached its task-specific saturation point, retaining additional representation dimensions imposes additional system cost without necessarily providing additional predictive value.

This is the central hypothesis that will be further tested using enzyme-function datasets, independent protein benchmarks, repeated random seeds, vector-search benchmarks, and direct infrastructure measurements.

12.1 Biological Holdout Difficulty

The effect of dimensionality cannot be separated from the biological structure of the benchmark.

The best observed class-level results across the hierarchical holdouts were:

Table 9: Best class-level results under different biological holdouts.

Holdout	Best D	Test Accuracy	Test Macro-F1
Family	768	0.9827	0.9798
Superfamily	1280	0.9282	0.8718
Fold	768	0.8900	0.7602

The same ESM-2 representation therefore exhibits substantially different performance depending on the biological separation between training and test proteins.

Increasing dimensionality does not eliminate this difference.

This suggests that dimensionality is not the only bottleneck in protein representation systems. Biological distribution shift can remain a dominant source of error even when the full 1280-dimensional representation is available.

12.2 Hierarchical Classification

The experiments also evaluated different biological label levels.

For example, family-level classification under the family-holdout setting achieved approximately:

- 96.23% test accuracy,
- 0.9146 test macro-F1,
- 0.9374 balanced accuracy.

The fold-level classification experiment under the fold-holdout setting was considerably more difficult, with the best observed result reaching approximately:

- 31.62% test accuracy,
- 0.1286 macro-F1,
- 0.1778 balanced accuracy.

These results reinforce the importance of evaluating dimensionality across multiple biological generalization regimes rather than relying on a single random train/test split.

13 Engineering Analysis

13.1 Storage Reduction

The most direct engineering benefit of dimensionality reduction is storage.

For float32 representations:

$$S = 4ND. \quad (26)$$

For a 100 million protein repository:

Table 10: Raw float32 embedding storage for 100 million proteins.

Dimensionality	Bytes per protein	Approx. Storage
1280	5,120	512 GB
1024	4,096	409.6 GB
768	3,072	307.2 GB
512	2,048	204.8 GB
256	1,024	102.4 GB
128	512	51.2 GB
64	256	25.6 GB
32	128	12.8 GB

The 256-dimensional representation therefore requires only 20% of the storage of the original 1280-dimensional representation.

At very large scale, this difference becomes operationally significant.

The reduction also applies to replicated storage. For example, with three copies of the embedding repository, the nominal raw storage difference between 1280D and 256D becomes:

$$3(512 - 102.4) = 1228.8 \text{ GB}. \quad (27)$$

This calculation excludes database indexes, metadata, compression, and replication overhead and should therefore be interpreted as a lower-level representation-storage estimate rather than a complete infrastructure cost model.

13.2 Memory and Batch Processing

Embedding dimensionality also determines tensor memory.

For a batch of B proteins:

$$M = BD \times 4. \quad (28)$$

For example, a batch of 100,000 embeddings requires:

$$100,000 \times 1280 \times 4 = 512 \text{ MB} \quad (29)$$

for the raw 1280-dimensional float32 matrix.

The corresponding 256-dimensional matrix requires:

$$100,000 \times 256 \times 4 = 102.4 \text{ MB}. \quad (30)$$

This can increase the amount of data that can be processed in memory without changing the biological model that generated the original representation.

13.3 Downstream Classifier Size

For a linear classifier with K classes, the number of parameters in the final layer scales as

$$P = DK + K. \quad (31)$$

Therefore, dimensionality reduction directly reduces the number of parameters required by downstream classifiers.

For nonlinear models, the effect can be substantially larger because the first hidden layer scales as

$$P_{\text{first}} = DH + H. \quad (32)$$

Thus, dimensionality reduction can reduce both model memory and the number of arithmetic operations required by downstream inference.

13.4 Vector Search and Retrieval

Protein representation systems increasingly use embeddings for similarity search and retrieval.

For a dense vector distance calculation, the basic arithmetic complexity is approximately

$$O(D). \quad (33)$$

Consequently, reducing the dimensionality from 1280 to 256 reduces the number of coordinate-level operations required for a direct distance calculation by a factor of five.

In an actual vector database, the end-to-end improvement will depend on the indexing algorithm, cache behaviour, hardware, quantization, and implementation.

Therefore, the present work establishes the dimensionality reduction opportunity but does not claim a five-fold end-to-end search speedup without direct benchmarking.

14 Discussion

The results provide evidence for a central engineering principle:

The largest representation available is not necessarily the most efficient representation required by a downstream system.

The 1280-dimensional ESM-2 representation contains substantial information that can be compressed without an immediate proportional loss in downstream classification performance.

For the class-level family-holdout experiment, performance increases rapidly up to approximately 256–512 dimensions and then enters a saturation regime.

This can be viewed as a representation efficiency curve:

$$\text{Utility}(D) = \frac{\text{Task Performance}(D)}{\text{Representation Cost}(D)}. \quad (34)$$

At very low dimensionality, the representation becomes too compressed and predictive performance deteriorates.

At very high dimensionality, additional representation capacity provides progressively smaller improvements while storage and computational costs continue to increase approximately linearly.

The engineering objective is therefore not simply to maximize predictive performance.

Instead, it is to identify the point at which additional dimensions no longer justify their operational cost.

14.1 The 256D Result

The 256-dimensional representation is particularly interesting from an engineering perspective.

It represents an 80% reduction in dimensionality:

$$1 - \frac{256}{1280} = 0.80. \quad (35)$$

At the same time, it retains 93.99% of PCA variance and achieves 97.64% test accuracy on the class-level family-holdout task.

The 512-dimensional representation improves accuracy to 98.11%, while 768 dimensions reaches 98.27%.

Thus, the choice between 256D, 512D, and 768D should depend on the acceptable performance tolerance of the application.

This is precisely why MERD should be defined relative to an explicit tolerance ϵ rather than declaring a universally optimal dimensionality.

For example, if an engineering application permits approximately a one percentage point reduction relative to the best observed accuracy, 512D may be sufficient for this task. If the tolerance is more stringent, a larger representation may be justified.

14.2 Why Full Dimensionality Can Be Wasteful

The experiment demonstrates an important asymmetry.

The cost of retaining additional dimensions is predictable:

$$C_{\text{representation}} \propto D. \quad (36)$$

However, the performance benefit is not linear:

$$\Delta P(D) \rightarrow 0 \quad (37)$$

as the representation approaches a task-specific saturation region.

This creates diminishing engineering returns.

For the evaluated task, increasing from 768D to 1280D increases representation storage by approximately

$$\frac{1280}{768} \approx 1.67 \quad (38)$$

times, while test accuracy remains approximately unchanged at 98.27%.

Thus, the additional dimensions have measurable representation variance but negligible observed classification benefit.

This is a key distinction between *representation completeness* and *engineering usefulness*.

14.3 Biological Generalization Changes the Optimal Dimension

The experiments also demonstrate that there is no universal MERD.

For the class-level family-holdout task, performance saturates around 768 dimensions.

For the superfamily-holdout task, the best observed result occurred at 1280 dimensions.

For the fold-holdout task, increasing dimensionality did not recover the performance observed in the easier family-holdout setting.

Therefore,

$$D^* = f(\text{task, generalization regime}). \quad (39)$$

This is important for engineering systems because dimensionality should be selected based on the actual workload.

A production protein search system, enzyme classifier, clustering system, and remote-homology system may have different effective dimensionality requirements.

15 Implications for Production Protein ML Systems

The results suggest a practical architecture for large-scale protein ML systems.

Instead of:

```
Protein
|
v
ESM-2
|
v
1280D embedding
|
+--> permanent storage
+--> vector database
+--> downstream models
+--> data transfer
```

a production pipeline can evaluate a task-specific compression layer:

```
Protein
|
v
ESM-2
|
v
```

```

1280D embedding
  |
  v
PCA / learned bottleneck
  |
  v
D-dimensional representation
  |
  +--> permanent storage
  +--> vector database
  +--> downstream models
  +--> retrieval

```

The high-dimensional representation can remain an intermediate artifact, while the lower-dimensional representation becomes the production representation.

This creates an important separation:

$$\text{Model representation} \neq \text{Production representation.} \quad (40)$$

A model may naturally produce a 1280-dimensional embedding while the downstream system only requires 256 or 512 dimensions.

16 Limitations

Several limitations should be considered.

16.1 Single Protein Language Model

The current study evaluates ESM-2 650M with a 1280-dimensional representation. The observed dimensionality behaviour may differ for smaller or larger protein language models.

ESM-2 itself is available at multiple model scales with different native embedding dimensions, demonstrating that representation dimensionality is already a design variable within the ESM family [1].

16.2 PCA Is Post-Hoc

The current compression occurs after ESM-2 inference.

Consequently, the results do not demonstrate a reduction in ESM-2 encoder compute.

The primary demonstrated benefits are downstream storage, memory, data movement, and downstream computation.

16.3 Classification-Centric Evaluation

The current results use logistic regression as the downstream classifier.

Other downstream models may exploit information differently, and the dimensionality-performance curve may therefore change.

16.4 Single Random Seed

The current dimensionality sweep uses a single random seed.

Small differences between neighbouring dimensionalities should therefore be validated using repeated seeds before being treated as statistically meaningful.

16.5 Benchmark Scope

The main results currently come from the TAPE Remote Homology benchmark.

Additional protein tasks are required to determine whether the observed saturation behaviour generalizes across enzyme function, localization, stability, and other biological prediction tasks.

17 Future Work

The next phase of the study will expand the engineering evaluation in several directions.

17.1 Enzyme Function Prediction

An enzyme-function benchmark based on hierarchical Enzyme Commission labels will be used to determine whether different levels of biological functional resolution require different representation dimensionalities.

The analysis will evaluate:

$$D \times \text{EC hierarchy level.} \quad (41)$$

This will test whether broad enzyme classes saturate at smaller dimensionalities than fine-grained enzyme functions.

17.2 Independent Protein Benchmark

An independent benchmark such as Pfam or a protein localization dataset will be evaluated to determine whether the dimensionality saturation behaviour generalizes beyond Remote Homology.

17.3 Repeated Seeds

Key dimensionalities will be repeated across multiple random seeds:

$$\{42, 123, 456, 789, 2024\}. \quad (42)$$

Mean performance and standard deviation will be reported.

17.4 Direct Infrastructure Benchmarking

The most important engineering extension is to measure actual system-level costs.

For each dimensionality, the following quantities should be measured:

- embedding storage size,
- RAM consumption,
- GPU memory consumption,
- serialization time,
- data-transfer volume,
- vector-index size,
- vector-search latency,
- vector-search throughput,

- downstream model size,
- downstream inference latency,
- batch throughput.

This will allow the study to move from theoretical scaling arguments to measured engineering performance.

17.5 Quantization

Dimensionality reduction should also be compared with numeric quantization:

$$1280D\text{-FP32} \rightarrow 256D\text{-FP32} \quad (43)$$

versus

$$1280D\text{-FP32} \rightarrow 1280D\text{-FP16/INT8}. \quad (44)$$

Combining dimensionality reduction and quantization may produce substantially larger infrastructure savings.

17.6 Learned Bottlenecks

A learned projection could be compared with PCA:

$$z = f_{\theta}(x), \quad x \in \mathbb{R}^{1280}, \quad z \in \mathbb{R}^D. \quad (45)$$

This would determine whether supervised dimensionality reduction can achieve better performance-cost trade-offs than unsupervised PCA.

17.7 Smaller Protein Language Models

To investigate encoder compute directly, smaller ESM models can be compared against ESM-2 650M.

For example, the ESM family includes models with native embedding dimensions substantially smaller than 1280 [1].

This enables a second engineering question:

Can we reduce not only the downstream representation cost, but also the cost of generating the representation?

18 Conclusion

This work investigated the engineering consequences of treating protein language model embedding dimensionality as an optimizable system parameter rather than a fixed property that must always be retained in full.

Using ESM-2 650M embeddings with a native dimensionality of 1280, we evaluated PCA representations from 32 to 1280 dimensions on the TAPE Remote Homology benchmark.

The results show a clear diminishing-return pattern. Class-level family-holdout accuracy increased from 83.73% at 32 dimensions to 97.64% at 256 dimensions and 98.11% at 512 dimensions. Performance reached 98.27% at 768 dimensions and remained approximately unchanged at 1024 and 1280 dimensions.

This result is particularly important because PCA explained variance continued increasing substantially over the same range. The 768-dimensional representation retained 99.10% of

variance, while 1280 dimensions retained 100%, yet both achieved approximately the same test accuracy.

Therefore, preserving the maximum possible representation variance is not necessarily the same as maximizing engineering value.

The results also show that the appropriate dimensionality depends on the biological generalization problem. More difficult holdout regimes remain challenging even when the full 1280-dimensional representation is retained. Consequently, dimensionality reduction should be viewed as a task-dependent optimization problem rather than a universal fixed setting.

From an engineering perspective, reducing 1280 dimensions to 256 dimensions provides an 80% reduction in vector dimensionality and a five-fold reduction in raw float32 embedding storage. At 100 million proteins, this corresponds to reducing the raw representation footprint from approximately 512 GB to 102.4 GB before indexing and metadata overhead.

The central engineering principle emerging from this work is therefore:

Use the smallest protein representation that satisfies the required downstream performance.

We propose the Minimum Effective Representation Dimension (MERD) as a practical framework for identifying this point.

The broader implication is that protein language model systems should distinguish between the dimensionality naturally produced by a foundation model and the dimensionality actually required by a production application. This distinction can enable substantial reductions in storage, memory, network traffic, vector-search cost, and downstream model complexity without necessarily sacrificing meaningful predictive performance.

Future work will extend the evaluation to enzyme-function prediction and independent protein benchmarks, repeat experiments across multiple seeds, and directly benchmark infrastructure-level costs. These experiments will determine whether the observed dimensionality-efficiency relationship can be generalized into a practical engineering methodology for large-scale protein AI systems.

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